NephroTwin: architecture of a personal kidney-function digital twin

DATA

Longitudinal records

Synthea synthetic EHR (CSV)
or any longitudinal cohort table

Future live feed

ABDM / FHIR R4 lab results (consented)

Ingestion and data quality

Outpatient creatinine only (acute-care values excluded)
Creatinine vs reported eGFR consistency check in the same encounter
Follow-up stops at kidney failure (dialysis or transplant) · isolated-spike removal
Pseudonymous IDs; names and identity numbers are never read

CLINICAL

Clinical engine

eGFR: CKD-EPI 2021 (race-free)
KDIGO 2024 G / A staging and risk band
KFRE 4-variable (2- and 5-year)

Population model

Hierarchical prior over level and slope
EM fit · shrinkage chosen by cross-validation
Dynamics tuned on predictive accuracy

Personal twin (per patient)

Bayesian state space: [eGFR, slope]
Robust Kalman update after every result
Split-conformal 80% intervals

SERVICES

Forecast and risk

eGFR trajectory with uncertainty
P(eGFR < 30), P(eGFR < 15), rapid decline
Atypical results flagged for repeat test

What-if simulator

SGLT2i · ACEi/ARB · GLP-1 RA · BP target
Trial-based effect ranges, Monte Carlo
Paired paths: usual care vs plan

Explanation

Groq LLM (openai/gpt-oss-120b)
Facts-only prompt, no dosing advice
Deterministic template fallback

USE

Streamlit dashboard

Replay results one at a time · what-if · plain-language summary
Clinician records a care plan; later results are compared with it

Validation report

Held-out test patients · 5 baselines · bootstrap 95% CIs
Interval coverage · event AUROC · learning curve

Feedback loop: new results and recorded plans update the twin